PADHAKU
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CHAPTER 4

Data Handling and Presentation

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡
answers are samples: write your own.

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⭐ 4.1 Collecting and Organising Data

? To figure out the most popular game in their class, what should Navya and Naresh do?

💡 SAMPLE ANSWER

Ask every student in the class about their favourite game and write down the answers (collect data).

Then count how many students chose each game. The game chosen by the most students is the most popular.

🧮 Figure it Out · 4.1 (Favourite games)

1 What would you do to find the most popular game among Naresh's and Navya's classmates?

💡 SAMPLE ANSWER

Organise the list in a table: write each game once and put a tally mark for every student who chose it.

Count the tally marks. The game with the largest count is the most popular.

2 What is the most popular game in their class?

✔ ANSWER

Count each game from the list:

Game	Number of students

Hockey	8
Kabaddi	6
Cricket	6
Satoliya (Pittu)	5
Football	4
Badminton	2

Total = 31 students.

The most popular game is **Hockey** (8 students).

3 **Try to find out the most popular game among your classmates.**

💡 **SAMPLE ANSWER**

Ask each classmate their favourite game, make a tally table, and count.

Example: Cricket 12, Kho-kho 8, Kabaddi 6, Badminton 4 → most popular is **Cricket**. (Your class data will be different.)

4 Put a tick (✓) where Pari needs to collect data and a cross (×) where she does not:

a. What is the most popular TV show among her classmates?

b. When did India get independence?

c. How much water is getting wasted in her locality?

d. What is the capital of India?

✔ ANSWER

a. ✓ – she must ask her classmates.

b. × – it is a known fact (15 August 1947).

c. ✓ – she must observe and measure in her locality.

d. × – it is a known fact (New Delhi).

Figure it Out · 4.1 (Shri Nilesh's sweets)

1 Complete the table (tally marks → number of students):

Sweet	Tally marks (groups of 5 + single marks)	No. of students
Jalebi	5 + 1	6
Gulab jamun	5 + 5	9
Gujiya	5 + 5 + 3	?
Barfi	3	?
Rasgulla	5 + 2	?

a. How many students chose jalebi? b. Barfi was chosen by __ students. c. How many chose gujiya? d. Rasgulla was chosen by __ students. e. How many chose gulab jamun?

✔ ANSWER

Each bundle of four lines with a line across = 5.

Gujiya: 5 + 5 + 3 = 13; Barfi: 3; Rasgulla: 5 + 2 = 7.

a. **6**

b. **3**

c. **13**

d. **7**

e. **9**

2 **Is the above table sufficient to distribute each type of sweet to the correct student? Explain. If it is not sufficient, what is the alternative?**

✔ ANSWER

No. The table only tells how many students chose each sweet, not which student chose which sweet.

Alternative: write each student's name with the sweet he/she chose (or list the names under each sweet).

Figure it Out · 4.1 (Shoe sizes)

1 **Shoe sizes in ascending order: 3, 3, 3, 4, 4, 4, 4, 4, 4, 4, 4, 4, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 6, 6, 6, 6, 7**

a. The largest shoe size is __. b. The smallest shoe size is __.

c. There are __ students who wear size 5. d. There are __ students who wear sizes larger than 4.

✔ ANSWER

a. Last number in the list → **7**

b. First number in the list → **3**

c. Count the 5s → **10** students

d. Sizes 5, 6 and 7: 10 + 4 + 1 = **15** students

2 **How did arranging the data in ascending order help to answer these questions?**

✔ ANSWER

The smallest value comes first and the largest comes last, so we see them at once.

Equal values come together, so counting how many students wear each size is quick and easy.

3 **Are there other ways to arrange the data?**

✔ ANSWER

Yes. We can arrange it in descending order, or make a frequency table:

Shoe size	No. of students
3	3
4	9
5	10
6	4
7	1

4 **Record the trees you see on your way from home to school in a table (Tree | No. of trees). a. Which tree was found in the greatest number? b. Which tree was found in the smallest number? c. Were any two trees found in the same numbers?**

💡 SAMPLE ANSWER

Sample data:

Tree	No. of trees
Peepal	4
Neem	7
Mango	3
Banyan	1
Ashoka	3

a. **Neem** (7)

b. **Banyan** (1)

c. **Yes**, Mango and Ashoka (3 each).

(Your answers depend on the trees you observe.)

5 **Paste a small news item. Count the letters ‘c’, ‘e’, ‘i’, ‘r’, ‘x’ (and one letter of your choice) and fill a table.**

a. The letter found the most number of times is __. b. The letter found the least number of times is __.

c. List c, e, i, r, x in ascending order of frequency. Is your order the same as your classmates’? (Most get ‘x, c, r, i, e’.) Why?

💡 **SAMPLE ANSWER**

Sample table:

Letter	c	e	i	r	x	t
Number of times	9	38	21	19	1	27

a. **e**

b. **x**

c. Order: **x, c, r, i, e**. Almost everyone gets the same (or nearly the same) order because in English words ‘e’ is used very often and ‘x’ is used very rarely. This pattern holds in almost any English text.

5 (d) **Write the process you followed to complete this task.**

💡 **SAMPLE ANSWER**

- Made a table with the letters c, e, i, r, x.

- Read the article word by word; for each of these letters put a tally mark in its column.
- Counted the tally marks and wrote the numbers.
- Compared the numbers to find the most and least frequent letters.

5 (e) **Discuss with your friends the processes they followed.**

💡 **SAMPLE ANSWER**

Some friends may count one letter at a time through the whole article; others may tick all five letters in one reading using tally marks. One reading with tally marks is faster; one letter at a time is less confusing but slower.

5 (f) **If you do this task with another news item, what process would you follow?**

💡 **SAMPLE ANSWER**

I would read the article once and put tally marks for all five letters together in a table, then count the tallies. Crossing out each word after checking it helps avoid missing or double-counting letters.

⭐ 4.2 Pictographs

? **(Modes of travel pictograph, 1 symbol = 1 student) Which mode of travel is used by the most number of students? Which is used by the least?**

💡 **SAMPLE ANSWER**

Look at the row with the most symbols → most used; the row with the fewest symbols → least used.

From the picture: most students come by **school bus**; the least come by **private car**. (check the picture in your book)

Example **Nand Kishor's pictograph (1 symbol = 10 children): Always – 5 symbols, Sometimes – 2½ symbols, Never – 4 symbols.**

1. How many children always slept at least 9 hours? 2. How many sometimes slept at least 9 hours? 3. How many always slept less than 9 hours?

✔ **ANSWER**

1. 5 × 10 = **50 children**

2. 2 × 10 = 20, half symbol = 5 → 20 + 5 = **25 children**

3. ‘Never’ has 4 symbols: 4 × 10 = **40 children** (they never sleep 9 hours, so they always sleep less).

? **Students present: I – 30, II – 35, III – 20, IV – 25, V – 30, VI – 25, VII – 30, VIII – 20.**
If they use one symbol for each student (like Lakhanpal), what challenges might they face?

✔ ANSWER

There are 215 students in all, so they would have to draw 215 symbols (up to 35 in one row).

It takes a lot of time and space, the rows become very long, and counting the symbols to read the graph is hard and error-prone.

Math Talk **If one symbol = 10 students and a half symbol = 5 students, what problems arise if the number of students present in a class is 33 or 27?**

✔ ANSWER

33 = 3 full symbols + 3 students; 27 = 2 full symbols + 5 (half) + 2 students.

The 3 or 2 extra students are less than 5, so we would have to draw small, odd parts of the symbol (like 3/10 of it). Such parts are hard to draw and hard to read correctly.

So pictographs are difficult when numbers are not multiples of the scale.

Figure it Out · 4.2 Pictographs

1 **Pictograph of books borrowed in a week from the library of Middle School, Ginnori (1 symbol = 1 book).**

a. On which day were the minimum number of books borrowed? b. What was the total number of books borrowed during the week? c. On which day were the maximum number of books borrowed? What may be the possible reason?

💡 SAMPLE ANSWER

a. **Thursday**

b. Add the symbols of all days → **24 books**

c. **Saturday**. Possible reason: Sunday is a holiday, so students borrow more books on Saturday to read over the weekend.

(check the picture in your book)

2 **Kites sold: Chaman 250, Rani 300, Rukhsana 100, Jasmeet 450, Jetha Lal 250, Poonam Ben 700. Make a pictograph with 1 symbol = 100 kites.**

a. How many symbols represent Rani's kites? b. Who purchased the maximum number of kites? c. Who purchased more, Jasmeet or

Chaman? d. Rukhsana says Poonam Ben purchased more than double Rani's kites. Is she correct? Why?

✔ ANSWER

Pictograph (★ = 100 kites, ½★ = 50 kites):

Shopkeeper	Kites
Chaman	★★½
Rani	★★★
Rukhsana	★
Jasmeet	★★★★½
Jetha Lal	★★½
Poonam Ben	★★★★★★★

a. 300 ÷ 100 = **3 symbols**

b. **Poonam Ben** (700 kites)

c. **Jasmeet** (450 > 250)

d. **Yes.** Double of Rani's = 2 × 300 = 600, and Poonam Ben bought 700 > 600.

⭐ 4.3 Bar Graphs

1 **(Bar graph of students absent) In Class 2, ___ students were absent that day.**

✔ ANSWER

5

2 **In which class were the maximum number of students absent?**

✔ ANSWER

Class 8 (7 students, the tallest bar)

3 **Which class had full attendance that day?**

✔ ANSWER

Class 5 (0 students absent, no bar)

🧮 Figure it Out · 4.3 Bar Graphs (Traffic)

1 **How many total cars passed through the crossing between 6 a.m. and noon?**

✔ ANSWER

Read each bar: 6–7: 150, 7–8: 1200, 8–9: 1000, 9–10: 800, 10–11: 700, 11–12: 600

Total = 150 + 1200 + 1000 + 800 + 700 + 600 = **4450 vehicles**

2 **Why do you think so little traffic occurred during 6–7 a.m., as compared to the other hours from 7 a.m. to noon?**

💡 SAMPLE ANSWER

It is very early in the morning. Most people have not yet left home; offices, shops and schools have not opened.

3 **Why do you think the traffic was the heaviest between 7–8 a.m.?**

💡 **SAMPLE ANSWER**

This is the rush hour: children go to school and many people leave for work at this time.

4 **Why do you think the traffic was lesser and lesser each hour after 8 a.m. all the way until noon?**

💡 **SAMPLE ANSWER**

Most people have already reached school or work, so fewer people need to travel as the morning goes on.

⭐ 4.3 Example: Population of India

? **Population of India (in crores): 1951 – 36, 1961 – 44, 1971 – 54, 1981 – 68, 1991 – 84, 2001 – 102.**

On the basis of this bar graph, what may be a few questions you may ask your friends?

💡 **SAMPLE ANSWER**

- In which year was the population the highest/lowest?

- In which decade did the population increase the most?
- By how much did the population increase from 1981 to 1991?
- In which year did the population first cross 50 crores? 100 crores?

? **How much did the population of India increase over 50 years? How much did the population increase in each decade?**

✔ ANSWER

Over 50 years: 102 − 36 = **66 crores**

Each decade:

Decade	Increase
1951–1961	44 − 36 = 8 crores
1961–1971	54 − 44 = 10 crores
1971–1981	68 − 54 = 14 crores
1981–1991	84 − 68 = 16 crores
1991–2001	102 − 84 = 18 crores

⭐ 4.4 Drawing a Bar Graph (Imran's family)

1 **On which item does Imran's family spend the most and the second most?**

✔ ANSWER

Most: **Food** (₹3400). Second most: **House rent** (₹3000).

2 **Is the cost of electricity about one-half the cost of education?**

✔ ANSWER

Half of ₹800 = ₹400, and electricity = ₹400.

Yes, it is exactly one-half.

3 **Is the cost of education less than one-fourth the cost of food?**

✔ ANSWER

One-fourth of ₹3400 = 3400 ÷ 4 = ₹850.

Education = ₹800 < ₹850.

Yes.

Figure it Out · 4.4 Drawing a Bar Graph

1 **Samantha's data: Mites 6, Caterpillars 10, Beetles 5, Butterflies 3, Grasshoppers 2. Help her prepare a bar graph.**

✔ ANSWER

Steps:

- Draw a horizontal line (insects) and a vertical line (number seen).
- Scale: 1 unit length = 1 insect; mark 0, 1, 2, ... 10 on the vertical line.
- Draw bars of equal width with equal gaps:

Insect	Height of bar
Mites	6 units

Caterpillars	10 units
Beetles	5 units
Butterflies	3 units
Grasshoppers	2 units

- Give the title: “Insects and critters seen in a tea garden”.

2 **Tickets sold: Vidisha 24, Jabalpur 20, Seoni 16, Indore 28, Sagar 16. Part of the bar graph is erased.**

a. Write tickets for Vidisha above the bar. b. Write tickets for Jabalpur above the bar.

c. Vidisha’s bar is 6 units and Jabalpur’s is 5 units. What is the scale? d. Draw the correct bar for Sagar.

e. Add the scale on the vertical axis. f. Are the bars for Seoni and Indore correct? If not, draw the correct bar(s).

💡 **SAMPLE ANSWER**

a. **24**

b. **20**

c. 24 ÷ 6 = 4 and 20 ÷ 5 = 4 → **1 unit length = 4 tickets**

d. Sagar: 16 ÷ 4 = **4 units** high.

e. Mark 0, 4, 8, 12, 16, 20, 24, 28 (one number for each unit, starting at 0).

f. Seoni needs 16 ÷ 4 = 4 units → its bar is **correct**. Indore needs 28 ÷ 4 = **7 units** → its bar is **not correct**; redraw it 7 units high. (check the picture in your book)

3 **Chinu listed vehicles passing his house from 9 a.m. to 10 a.m. (50 entries: bike, car, bus, bicycle, auto rickshaw, scooter, bullock cart...).**

a. Prepare a frequency distribution table. b. Which means of transport was used the most? c. If you were collecting this data, how would you do it?

✔ ANSWER

a.

Means of transport	Frequency
Bike	13
Scooter	9
Bicycle	8
Auto rickshaw	8
Car	6
Bus	4
Bullock cart	2
Total	50

(Count with tally marks: one line per vehicle, every 5th line drawn across the previous four.)

b. **Bike** (13 times)

c. - Make a table with vehicle names in one column and tallies in the next.

- Sit at the roadside from 9 to 10 a.m. with a watch.
- Put a tally mark for each vehicle as it passes (add a new row for a new type).
- At the end, count the tallies and write the frequencies.

4 **Roll a die 30 times and record the numbers. Make a frequency table using tally marks. Find the number that appeared:**

a. the minimum number of times b. the maximum number of times c. numbers that appeared an equal number of times

💡 **SAMPLE ANSWER**

Sample result:

Number	Frequency
1	4
2	6
3	5
4	7
5	4
6	4
Total	30

a. **1, 5 and 6** (4 times each)

b. **4** (7 times)

c. **1, 5 and 6** appeared equally (4 times).

(Your results will be different because each roll is by chance.)

5 **Wickets taken by Jaspreet Bumrah in his last 30 matches:**

Wickets taken	Number of matches
0	2
1	4
2	6
3	8
4	3
5	5
6	1
7	1

a. What information is this table giving? b. What may be the title? c. What caught your attention? d. In how many matches has Bumrah taken 4 wickets?

✔ ANSWER

a. It tells in how many matches Bumrah took 0, 1, 2, ... 7 wickets.

b. “Wickets taken by Jaspreet Bumrah in his last 30 matches”

c. (Sample) Most often (8 matches) he took 3 wickets; he took 7 wickets in one match; only in 2 matches he took no wicket.

d. **3 matches**

5 (e) **Mayank says, “To know the total wickets in the last 30 matches, we add 0, 1, 2, 3, ... up to 7.” Can he get the total this way? Why?**

✔ ANSWER

No. Each number of wickets happened in many matches, not just once.

Example: 3 wickets were taken in 8 matches, which is 3 × 8 = 24 wickets, not just 3.

5 (f) **How would you correctly find the total number of wickets using this table?**

✔ ANSWER

Multiply wickets × number of matches for each row and add:

0×2 + 1×4 + 2×6 + 3×8 + 4×3 + 5×5 + 6×1 + 7×1

= 0 + 4 + 12 + 24 + 12 + 25 + 6 + 7

= **90 wickets**

6 **Pictograph of tractors in five villages (1 symbol = 1 tractor).**

a. Which village has the smallest number of tractors? b. Which village has the most? c. How many more tractors does Village C have than Village B? d. Komal says, "Village D has half the number of tractors as Village E." Is she right?

💡 SAMPLE ANSWER

a. **Village D**

b. **Village C**

c. **3** more tractors

d. **Yes**, D's symbols are half of E's.

(check the picture in your book)

7 **Pictograph of girl students in Classes 1–8 (1 symbol = 4 girls).**

a. Which class has the least girls? b. Difference between girls in Classes 5 and 6? c. If two more girls join Class 2, how would the graph change? d. How many girls are in Class 7?

💡 SAMPLE ANSWER

a. **Class 8**

b. **6** girls

c. 2 girls = half a symbol. Class 2 ends with a half symbol, so it becomes a **full symbol** (one more complete symbol).

d. 3 symbols × 4 = **12 girls**

(check the picture in your book)

8 **Mudhol dogs: Village A 18, B 36, C 12, D 48, E 18, F 24. Prepare a pictograph.**

a. What will be a useful scale or key? b. How many symbols for Village B? c. Kamini says dogs in B and D together are more than in the other 4 villages. Is she right?

✔ ANSWER

a. All numbers are multiples of 6, so **1 symbol = 6 dogs**.

Pictograph (🐕 = 6 dogs):

Village	Dogs
A	🐕🐕🐕
B	🐕🐕🐕🐕🐕🐕
C	🐕🐕
D	🐕🐕🐕🐕🐕🐕🐕🐕
E	🐕🐕🐕
F	🐕🐕🐕🐕

b. 36 ÷ 6 = **6 symbols**

c. B + D = 36 + 48 = 84; A + C + E + F = 18 + 12 + 18 + 24 = 72.

84 > 72, so **Yes, Kamini is right**.

9 **Survey of 120 students: Playing 45, Reading story books 30, Watching TV 20, Listening to music 10, Painting 15.**

Draw a bar graph with 1 unit length = 5 students. Which activity is preferred by most students other than playing?

✔ ANSWER

Heights of bars (divide by 5):

Activity	Height
Playing	45 ÷ 5 = 9 units
Reading story books	30 ÷ 5 = 6 units
Watching TV	20 ÷ 5 = 4 units
Listening to music	10 ÷ 5 = 2 units
Painting	15 ÷ 5 = 3 units

Mark 0, 5, 10, ... 45 on the vertical axis and draw bars of equal width with equal gaps.

Other than playing, most students prefer **reading story books** (30).

10 **Bar graph of saplings planted in the first week of July (Monday to Sunday).**
a. Total saplings planted on Wednesday and Thursday is __. b. Total saplings in the whole week is __.
c. Greatest number planted on __ and least on __. Why? How could you check your explanation?

💡 SAMPLE ANSWER

a. **70**

b. **310**

c. Greatest on **Saturday**, least on **Wednesday**.

Possible reasons: rain on some days; more students/teachers free on Saturday; on some days fewer people were present.

To check: look at the weather record and the attendance for each day, or ask the teachers and students.

(check the picture in your book)

11 **Number of tigers in India (approx.): 2006 – 1400, 2010 – 1700, 2014 – 2200, 2018 – 3000, 2022 – 3700. Their bar graph (scale 0, 1000, 2000, 3000, 4000) has mistakes. Find and fix them.**

💡 SAMPLE ANSWER

Compare each bar with the table. The bars for **2006, 2010, 2014 and 2018** are drawn wrong; only 2022 matches.

Correct lengths on the scale (1000 per mark):

Year	Bar should end at
2006	1400 (a little less than half-way from 1000 to 2000)
2010	1700
2014	2200 (just past 2000)
2018	3000 (exactly at 3000)
2022	3700

Redraw these bars to the correct lengths.

(check the picture in your book)

⭐ 4.5 Artistic and Aesthetic Considerations

? **Tallest mountains: Everest 8848 m, Aconcagua 6962 m, Denali 6194 m, Kilimanjaro 5895 m, Elbrus 5642 m, Vinson Massif 4892 m, Koscuiszko 2228 m.**

How much taller is Mount Everest than Mount Koscuiszko?

✔ ANSWER

8848 − 2228 = **6620 m** taller

? **Are Mount Denali and Mount Kilimanjaro very different in height?**

✔ ANSWER

6194 − 5895 = 299 m.

No, they differ by only about 300 m, so their heights are quite close.

🧮 Figure it Out · 4.5

1 **If you wanted to visually represent the heights of the tallest persons in each class in your school, would you use vertical bars or horizontal bars? Why?**

✔ ANSWER

Vertical bars. Height is measured upwards from the ground, so bars standing up like people make the comparison natural and easy to see.

2 **For a table of the longest rivers on each continent and their lengths, would you prefer vertical or horizontal bars? Why? Make the table and bar graph. Which continents have the longest rivers?**

💡 **SAMPLE ANSWER**

Horizontal bars, because a river's length lies along the ground, so horizontal bars look like rivers.

Approximate lengths:

Continent	Longest river	Length (km, approx.)
Africa	Nile	6650
South America	Amazon	6400
Asia	Yangtze	6300
North America	Mississippi–Missouri	6275
Europe	Volga	3530
Australia	Murray	2500

(Antarctica has no big river; it is covered with ice.)

Draw horizontal bars with scale 1 unit = 1000 km.

Longest rivers: **Africa (Nile)** and **South America (Amazon)**.

⭐ Infographics

? **In the triangle infographic, the taller triangles are also wider. Are taller mountains always wider?**

✔ ANSWER

No. A tall mountain can be narrow. The extra width gives wrong information, so the picture can mislead.

? **Is the mountain-range infographic better than the column graph with equal-width columns? The mountains look more realistic, but is the picture accurate?**

💡 SAMPLE ANSWER

It looks nicer, but it is **not accurate**. The shapes and widths are not to scale, so heights cannot be compared correctly. The column graph with equal widths is more accurate for comparing heights.

? **Everest appears to be twice as tall as Elbrus. What is 5642 × 2?**

✔ ANSWER

5642 × 2 = **11284**

> Everest is 8848 m, which is much less than 11284 m. So Everest is **not** twice as tall as Elbrus; the picture is misleading.

🎉 **Chapter 4 done! Revise the answers once more tomorrow.**